XXXX

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Education

TBD — Dec 2025 – May 2027
Bachelor of Science, Computer Science
- Transfer applicant (Fall 2026 intake) — decisions pending

CCC — Aug 2023 – Dec 2025
Associates of Science, Computer Science — *GPA: 3.7*
- Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Distributed Systems, High-Performance Computing, C++ Programming, Java Programming, Linear Algebra, Discrete Math, Differential Equations

Experience

Construction company — Nov 2025 – Feb 2026
Software Engineering Intern
- Shipped my software to six internal teams by drafting and presenting a technical design document to management
- Dropped IT JIRA ticket resolution time by 68% by owning end-to-end development of a C#/.NET tool
- Cut $53/month cloud costs by integrating an on-device inference engine using TinyLlama 1.1B (GGUF)
- Eliminated 100% of instance conflicts across all releases using an OS mutex, ensuring singleton execution
- Buffered 250+ events during file contention by leveraging an in-memory queue to cache logs until file was released
- Streamlined download time from 30s to 1s by shipping a self-contained installer wizard (with GUI) with optional configs
- Cut release time from 5m to 3s by using CI releases on tags to deliver one-command reproducible binaries and updates

Projects

Bang Programming Language | *Python, C++, Git, pytest, GitHub Actions*
- Created an interpreted programming language from scratch
- Decreased interpretation time 66% by optimizing hot execution paths using cProfile and line_profiler
- Lowered heap usage 50% by implementing compiler optimizations like constant folding, dead-code elimination, etc.
- Maintained 100% CI pass rate by automating 600+ unit/integration tests (94% code coverage) via GitHub Actions.

NextTube | *Python, JavaScript, FastAPI, Redis, PostgreSQL, AWS (S3), Docker*
- Built a local streaming service with FastAPI, Next.js, Postgres, Redis, and S3 orchestrated via Docker Compose
- Reduced abandonment 8% by implementing adaptive bitrate streaming via FFmpeg and hls.js
- Achieved near-linear scalability by horizontally scaling FFmpeg workers using Redis queues and Postgres concurrency
- Streamlined developer onboarding by consolidating 20+ CLI commands into a Docker Compose workflow

Notery | *Go, Gin, PostgreSQL, Redis, Meilisearch, Docker, Cloudflare R2, SvelteKit, JWT*
- Architected notes marketplace using Go/Gin backend, SvelteKit, Postgres, and Cloudflare R2
- Enabled sub-second feed queries for user-facing hot feeds by leveraging Redis ZSETs and optimal ranking algorithms
- Secured 100% of pay-walled assets by storing them in a secure R2 bucket with a server-side proxy
- Maintained data security by implementing RBAC through stateless JWT auth middleware and protected API groups

Chess Game | *Assembly, Irvine32*
- Developed a colored, playable Chess game in 32-bit Assembly (MASM)
- Dynamically rendered board graphics per turn with Windows system calls and Irvine32 library

Technical Skills

Languages: Python, Go, C#, JavaScript, C++, Assembly
Frameworks and Libraries: FastAPI, Gin, SvelteKit, Next.js, Pytest, HLS.js, NumPy, Irvine32
Tools: Docker, AWS, Git, GitHub Actions, PostgreSQL, SQLite, Redis, Meilisearch, Cloudflare R2, JWT
Concepts: Concurrency, REST API, Authentication, RBAC, Caching, Relational data modeling, Background processing, Job queues, CI/CD, Containerization, Performance profiling, Secure content delivery